

HOW QUANTUM COMPUTERS ARE SHATTERING TRADITIONAL ENCRYPTION PARADIGMS

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At its core, Quantum physics is the study of matter and energy at the most fundamental level. Classical physics deals with the macroscopic world that we see around us, but quantum physics reveals a whole other universe which goes against our everyday intuition, spanning topics of wave-particle duality, superposition, and quantum entanglement.

Then the question arises, what is quantum computing? A bit – whether classical or quantum- is a single unit of information. A classical bit can represent either 0 or 1, however quantum bits can exist in a state of superposition, meaning they can represent both 0 and 1 simultaneously. A good analogy to think about it is that of a spinning coin. While the coin is spinning, in a way it's heads and tails at the same time until you stop it.

When we try and measure this quantum bit, we end up forcing the qubit out of its superposition state and we end up with a definite state – either 0 or 1 – this process is called wave function collapse. In the context of our analogy, this is similar to collapsing the coin and observing the result – heads or tails. Furthermore, as dictated by the no-cloning theorem, it is impossible to create an identical clone of such a qubit in superposition – a principle that comes in very handy when dealing with quantum cryptography!

All this behaviour is comparable to that of waves. Just like waves in the ocean in the macroscopic world, waves in the quantum world can overlap and interfere, the associated probabilities with quantum states can also add up which can lead to interference patterns. These properties present qubits with a significant advantage that allows them to explore multiple possibilities at once and thus compute numerous calculations simultaneously – something that proves very dangerous to our current methods of encryption.

BREAKING CLASSICAL ENCRYPTION WITH QUANTUM COMPUTING

Right now, classical computers aren't a threat to modern encryption. Elliptical Curve Diffie-Hellman (ECDH) Key Exchange is a cryptographic protocol which allows 2 parties, commonly referred to as Alice and Bob, to securely communicate over a public channel by deriving a shared secret key, which they then use to encrypt their messages. Similar to standard Diffie-Hellman protocol, it uses asymmetric encryption features, over a public channel, the main difference being the hardness of the Discrete Logarithm Problem (DLP) vs the Elliptic Curve Discrete Logarithm Problem (ECDLP), which we come to later. We read a publication from Martin Kleppman¹, which helped us truly understand how quantum computers could be used to solve ECDLP. To demonstrate the workings of ECDH, we present the following scenario:

1. Both parties agree on a base point P , on an agreed elliptic curve (defined over a finite field) - this is public information. (The set of points on the elliptic curve, alongside the "point at infinity", forms an abelian group (operation being point additions/scalar multiplications) - this is important for later). Also, P acts as a generator for the large cyclic group.
2. Each party generates a private key for themselves - we'll call Alice's ' a ', and Bob's ' b '. (These are chosen from the set of integers Z^+ , up to ((the order of the base point P) - 1), which we'll call n)
3. They then calculate their public keys by performing the point addition operation k times (where k is their private key) - creating points in the form $A = aP$ and $B = bP$, which they make public when they exchange these points over their public channel.
4. To derive their mutual key i.e. same point on the curve, they both perform the point addition operation again k times on their private key, where k is the public key received from the other party e.g. in Alice's case, computing $a(bP) = abP$, and in Bob's, $b(aP) = baP$. These may seem different, but as mentioned before, the point addition operation is commutative - as the points on the elliptic curve is an abelian group. (Commutative operation is an abelian property).

The main reason this method of cryptography is secure, is based on the ECDLP assumption: "Given an elliptic curve E defined over a finite field F , a base point P on E , and a point Q on E such that $Q = kP$ for some integer k , it is computationally infeasible to determine k from P and Q ". Similar to the Decisional Diffie-Hellman Assumption (which I won't write here for space), these

¹ Martin Kleppmann. 2021. Implementing Curve25519/X25519: A Tutorial on Elliptic Curve Cryptography. 1, 1 (June 2021). - <https://www.cl.cam.ac.uk/teaching/2122/Crypto/curve25519.pdf> [Accessed 12th July 2024].

are currently computationally infeasible with today's technology. However, the DLP can be solved in sub-exponential time by using the index calculus method, which requires a larger amount of bit keys (typically 2048), to stay secure. This is an issue in modern consumer technology, such as smartphones, where minimising computational resources and storage requirements is crucial. As a result, modern elliptic curve cryptography is preferred, providing the same security with much smaller bits e.g. 256. The generic nature of elliptical curve groups (thought to have no exploitable structure other than the group law), means the best classical algorithm to solve ECDLP is the Pollard's rho algorithms for logarithms - which has a time complexity of $O(\sqrt{n})$. It seems we're safe - for now. This is where the threat of quantum computing enters. Even in our most secure cryptographic algorithms like ECDH, we still can't be sure of the safety of our data in the near future.

THE THREAT OF QUANTUM COMPUTING ON ELLIPTICAL CURVE DIFFIE-HELLMAN

Using "Algorithms for quantum computation: Discrete logarithms and factoring" by Peter Shor himself as a reference, we can adapt Shor's algorithm, which was built originally for finding discrete logarithms and prime factorisations (directly impacting the DLP, and also the RSA algorithm), to solve the ECDLP. Shor's algorithm works by using quantum parallelism to find the prime factors of a large number in polynomial time ($O((\log L)^3)$), where L is the number of bits). Shor's Algorithm works as follows:

We have 2 points P and Q , with an Elliptic curve E . We want to find k in $Q=kP$, (which is analogous to finding the value of a in $a(bp)=abp$, in the example with Alice and Bob from before):

1. We prepare an initial quantum state using three registers: two for representing coefficients a and b , and one for holding the elliptic curve point. The size of the quantum registers, which we'll call n , is calculated by rounding up $\log_2(\text{the number of elements in the elliptic curve})$. The initial state therefore is in the form $|0\rangle|0\rangle|0\rangle$, with the first 2 using n -qubits, and the third using $2n$ -qubits (as we need to store both x and y coordinates of elliptic curve points).
2. Then, we perform a Hadamard transformation on the first 2 registers in order to create a superposition of all of their (possible) states.

$$1/2^n \sum_{a=0, b=0}^{2^{n-1}} |a\rangle|b\rangle|0\rangle$$

3. We then use a quantum circuit to compute $aP + bQ$, and store the result in the third register, now making the state:

$$1/2^n \sum_{a=0, b=0}^{2^{n-1}} |a\rangle|b\rangle|(aP + bQ) \bmod (\text{the number of elements in the elliptic curve})\rangle$$

From here on, Quantum Fourier transformations are applied to the first 2 registers. The QFT is a crucial step as it helps identify the period of the function $aP + bQ$, which is essential because it relates to the hidden integer k in the ECDLP. After applying the QFT, we measure the state, obtaining information that can be processed classically to determine k - the one thing ECDH seeks to protect. (Unfortunately, due to space constraints, we can't dive into the details of the unitary transformations and other techniques used in Shor's algorithm.). And there we have it - we've just broken ECDH using quantum computers.

We've seen how quantum computing poses a significant threat to the security of cryptographic protocols like ECDH, potentially compromising today's secrets in the future. But don't start deleting your emails just yet! -NIST is developing quantum-resistant cryptographic protocols to protect today's sensitive information in the coming quantum era. While quantum computing is a risk to our data, it also opens new methods to distribute keys based on the fundamental laws of quantum physics, rather than a digital stream of bits powered by numerical sums.

SECURING OUR DATA WITH QUANTUM MECHANICS

One age-old innovation in secure communication comes from the 80s and introduces the idea of Quantum Key Distribution (QKD). The BB84 protocol, which we read up on in a publication from Nilanjana Datta², harnesses the power of quantum physics with simple instruments, whilst also proposing solutions to eavesdropping and mathematically based attacks from emerging technologies, such as quantum computers.

The protocol relies on the polarisation of photons to secure a key which can be used with a one-time pad. The beauty of this lies in the fact that the key is almost magically formed despite communications being completely public and open. A basic implementation of this protocol is as follows:

1. Alice sends Bob a series of photons, each randomly polarised in one of four possible ways (-, |, /, \), stemming from two main polarising bases – rectilinear (-, |) and diagonal (/, \). Note that each orientation within these bases represents a binary 1 or 0 and all we're concerned with is which state the photon is in relative to the respective base (rather than comparing 4 states independently).
2. Bob, not knowing the polarisations, randomly selects a basis to measure each photon – if he guesses right, he'll know the state of the photon, and consequently its binary representation. If he guesses wrong, the photon is essentially in superposition relative to that basis and there's a 50% chance it collapses into the wrong state.
3. Bob publicly announces which bases he used, and Alice confirms which are correct. Anyone eavesdropping knows the bases, but not the actual 'values', per se.
4. They 'sift' out the results in which Bob used an incorrect basis, ensuring they both agree on a common set of bits. These are then used as a key for communications.

And that's it, it's that simple – at least on paper. Bob and Alice have now produced a shared private key over public communications that they can use with the one-time pad method – each bit of plaintext is simply XOR'd with the corresponding bit of the key they've just produced. Though there is the element of randomness from which bits are discarded from the key, this step can only be wholly secure if the key itself is truly randomly generated – something that isn't possible with today's pseudorandom generators. However, Alice can leverage the inherent randomness of qubits in superposition to yield a truly random string of polarised photons. That is, if she had access to a suitable quantum system with appropriate Hadamard gates, of course, but I digress.

The elegance of utilising quantum mechanics lies in the fact that an eavesdropper can easily be detected simply by their measurements of the data. You see, when Alice transmits the photons, they are in a state of superposition, and as soon as Eve tries to measure them, they collapse into what is equally likely to be the intended form or an incorrect form. Once Bob receives this 'faulty' data he will check a random subset of the key publicly with Alice and they will inevitably find more errors than anticipated – in more technical terms, if the quantum bit error rate (QBER) is above a certain threshold, they can be certain their conversation has been eavesdropped.

IMPROVING SECURITY THROUGH PHOTON ENTANGLEMENT

Taking a step further into quantum cryptography, we come to a more direct, keyless protocol that is almost protected by the laws of quantum physics itself. Quantum Secure Direct Communication (QSDC) allows Alice and Bob to communicate using entangled photon pairs in which they embed their message in the particles' Bell states.

To understand more about Bell states, we first must establish what entanglement means. Quantum entanglement describes the interlinking of two or more particles such that the quantum state of each particle cannot be described independently of the state of the others. Consider two photons created in a lab that have been entangled, if we measure the polarisation of one to be 'vertical', it will follow that the other photon, no matter how far away, will immediately be found to have a corresponding polarisation. These particles could be separated by the breadth of the universe and still share the same link!

² Nilanjana Datta, 2020. *Quantum Information and Computation, Part IIC*. University of Cambridge. Available at: <https://www.qi.damtp.cam.ac.uk/files/PartIIC/QIC-6.pdf> [Accessed 16th July 2024].

Now, to determine the polarisation of the second photon based solely on the first, we have to look at Bell states. There are four Bell states which exhibit different correlations between entangled particles which can't be explained by classical physics. They are demonstrated through the following expressions:

$$\begin{aligned} |\Phi^+\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) & |\Psi^+\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \\ |\Phi^-\rangle &= \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) & |\Psi^-\rangle &= \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \end{aligned}$$

The two Φ states simply indicate that the qubits will be in the same state (00 or 11), with the negative showing a phase difference between them. The Ψ states work similarly, the only difference being that the qubits are in opposing states (01 or 10).

Of course, to transmit information we have to have some way of manipulating these states – this can be done with the following Pauli operators:

- *I* - Identity; Leaves the state as is
- *Z* - Pauli-Z; Flips the phase of the $|1\rangle$ component
- *X* - Pauli-X; Flips the state i.e. from $|1\rangle$ to $|0\rangle$ and vice versa
- *XZ* - Pauli-X then Pauli-Z; Flips both the state and the phase

In actuality, Alice encodes information in these operations and not in the Bell states themselves, this allows her to securely send not just one, but two 'conventional' bits worth of information per photon (4 quantum operations map to 2^2 bit combinations). With the protocol directly transferring information in the photons, i.e. without needing to establish a key, it is a much more efficient approach to quantum cryptography, not to mention the increased security against eavesdroppers due to its inherent reliance on entanglement. To better illustrate this, we present the following scenario:

1. Alice and Bob share pairs of entangled photons in a pre-decided Bell state; let's say $|\Phi^+\rangle$
2. Alice can then perform one of the four quantum operations to her photon. Here, she applies *Z* and flips the phase, changing it to $|\Phi^-\rangle$.
3. She then sends her photons over to Bob and Bob performs Bell state measurements on the pairs, identifying the new Bell state as $|\Phi^-\rangle$.
4. Knowing the initial and current Bell state, Bob can determine the operation applied to be *Z*. Subsequently, he now has the binary value Alice sent e.g. 01.
5. Publicly, Alice and Bob compare a subset of their data and determine the QBER, making them aware of any potential eavesdropping.

Though there are some things that have to be agreed upon before communication can occur, such as initial bell state of photon pairs and which quantum operation maps to which combination of 2 bits, these being decrypted isn't such an issue as Eve's presence can be detected in real time. See, at any given time either Alice or Bob (or both) possess entangled photons, meaning if Eve were to intercept and measure the photons (causing the photons to collapse), someone would be aware because the state of their entangled photons would collapse too, a phenomenon Einstein referred to as "spooky action at a distance". This adds to the fact that even if Eve somehow went undetected initially, she would be found through a heightened QBER, as seen with BB84.

Based on these discussions it almost sounds as if quantum methods of encryption are our saving grace, but there are many complexities that remain. For one, we can't yet manipulate individual photons and achieve an accuracy required to perfectly execute these methods, and we must send information in bursts of photons instead. This poses its own issues and introduces many loopholes to these methodologies, which opens a whole other conversation on protecting the transmission of photons. So, for now, quantum encryption isn't a perfect solution, but as we get closer to the inevitable demise of classical encryption algorithms, we can only hope science brings us nearer to the technology needed to successfully carry out these quantum methods of encryption.

Despite our hard work, we do still have curiosities about the future. We'd be honoured to visit Gonville & Caius College, and learn from the experts to satisfy our questions about where the world is going with quantum computing – is it the changes in cryptography we're all expecting, maybe breakthroughs in search with Grover's algorithm, or perhaps it takes an unexpected route in medicine? Regardless, thanks to Dr Timothy Jones for creating this opportunity and thank you for the taking the time to read our paper!